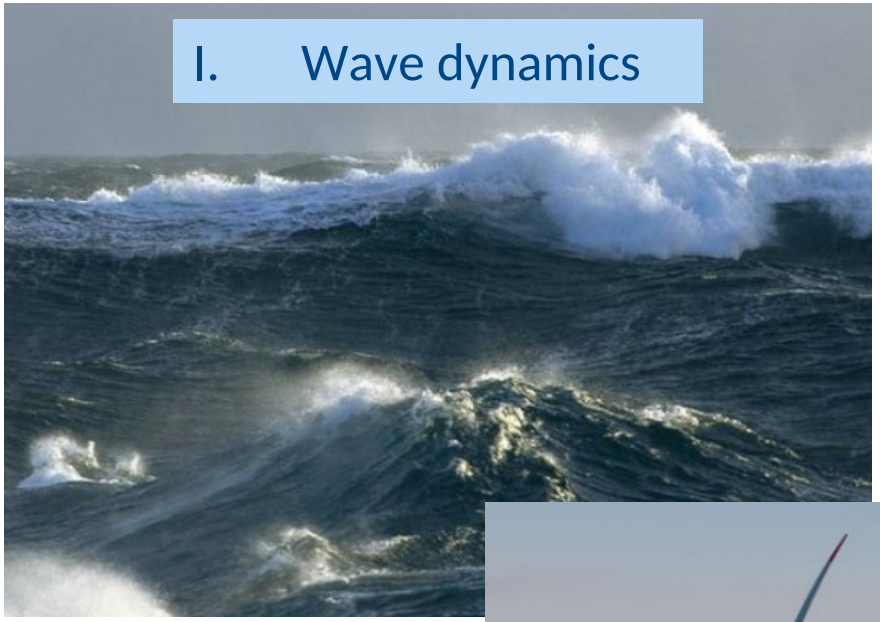
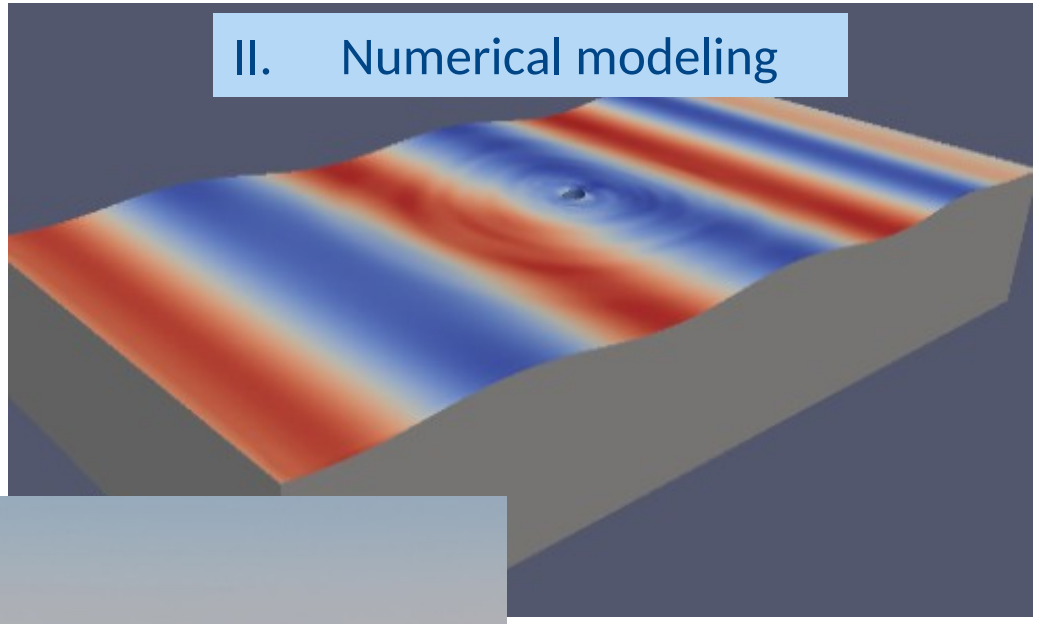
SSWPOWE Class project







Class project

Introduction and site selection (1 class)

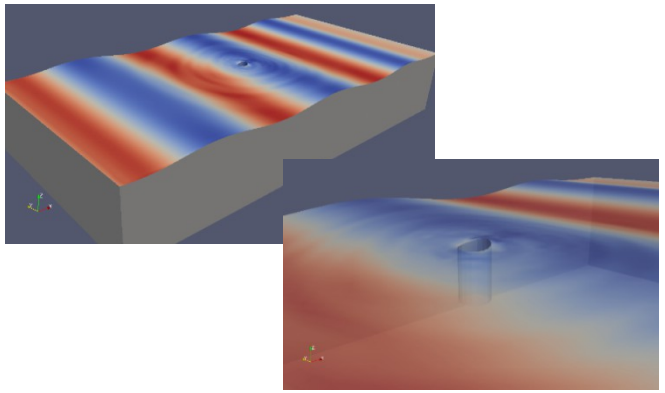
I. Wave dynamics (2 classes)

- Characterizing the wave characteristics at the selected study site and estimating theextreme wave conditions;



II. Numerical modeling (2 classes)

- Applying a numerical model to propagate waves from deep to shallow water conditions to estimate the wave forces on a structure;



III. Wave-structure interactions (2 classes)

- Estimating the movement of the wave turbine as well as the design criteria for the anchoring system.

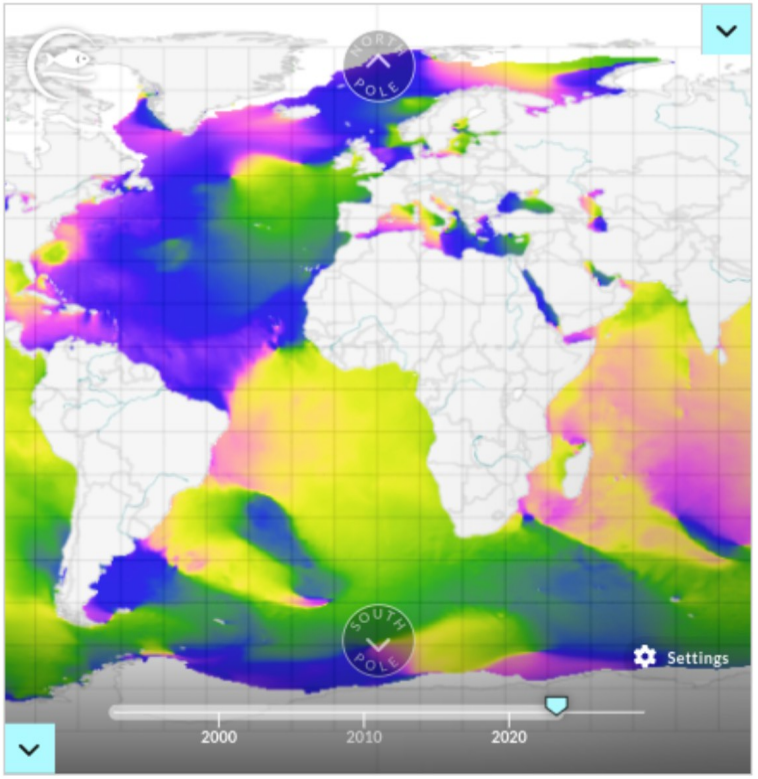


Finish analysis and prepare presentation (1 class)

Wave analyses….

Availability of long-term wave simulations via Copernicus:

https://marine.copernicus.eu/



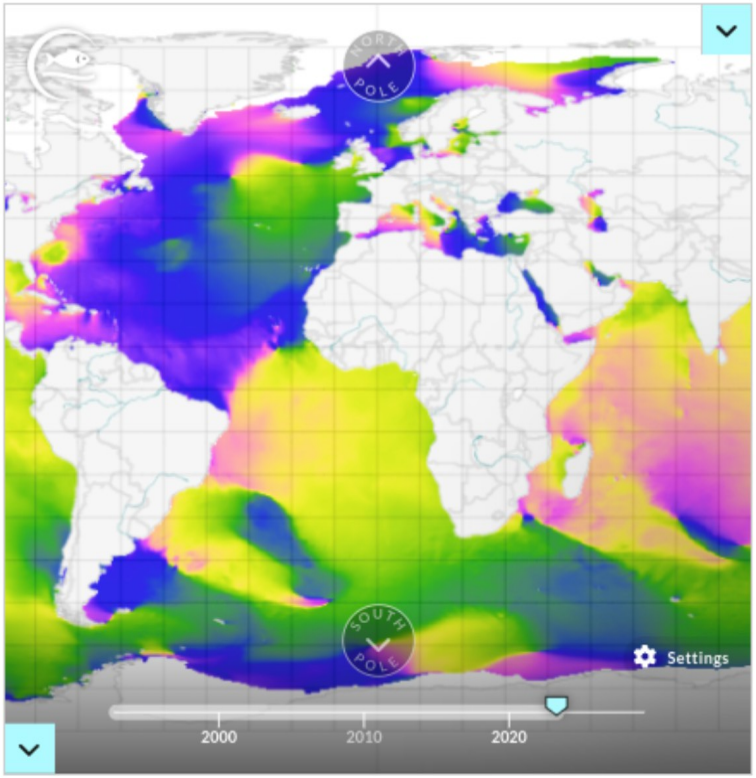
Class P2 (09/10): wave climate analysis

- Hs, T02, Dir: time series, histograms,
- Seasonality
- (if time: compare simulations to wave buoy observations)

Wave analyses….

Availability of long-term wave simulations via Copernicus:

https://marine.copernicus.eu/



Today class P3 (10/16): extreme value analysis of the wave height

- Estimate the 1- and 50-year return period wave heights?

Design load cases

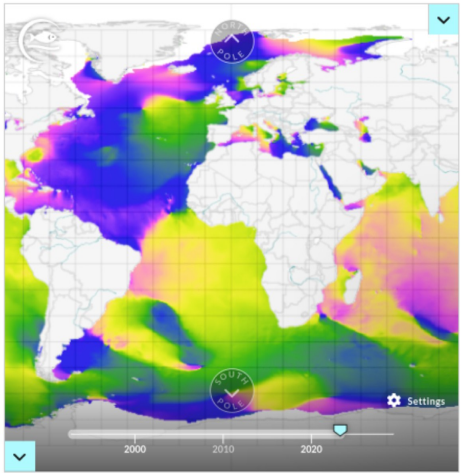
Design of a marine structure requires knowledge of extreme wave characteristics : develop design load cases (DLCs) to test the structural load response for:



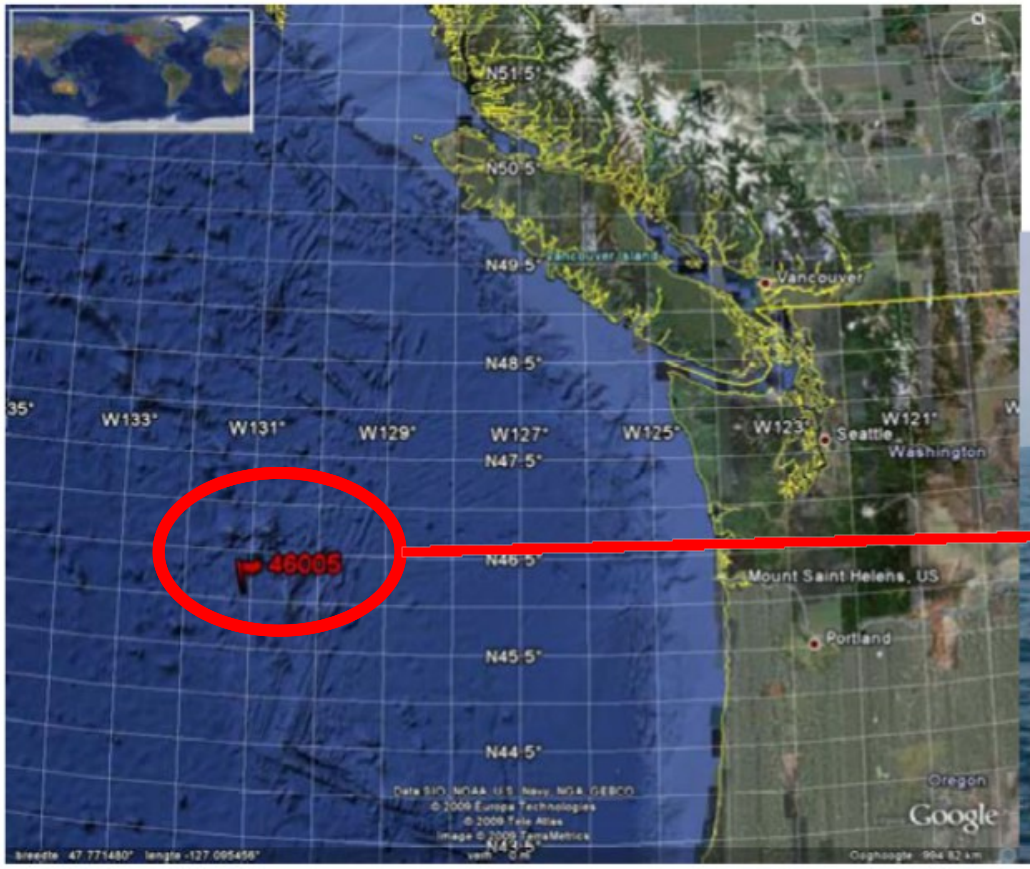
- Transportation
- Operation under normal conditions (occurring every year)
- Survival under extreme or abnormal environmental conditions for a range of return periods depending on the structure (e.g. 1, 5, 50, 100 years)

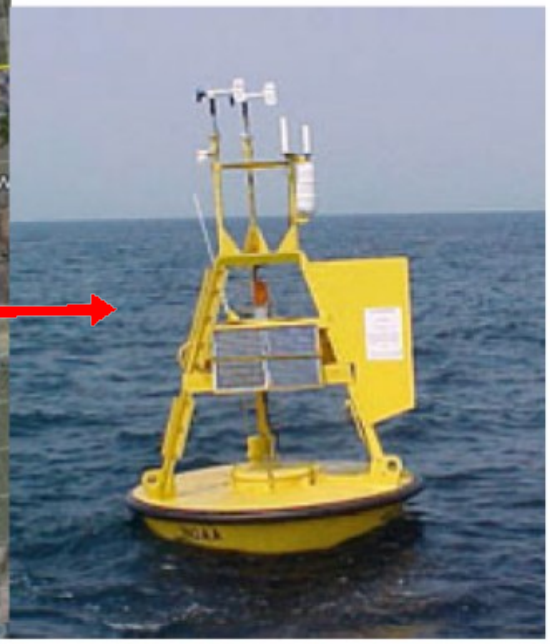
Extreme wave design

1. Obtaining wave data at the desired location of sufficient duration



(Caires, 2011, JCOMM report)



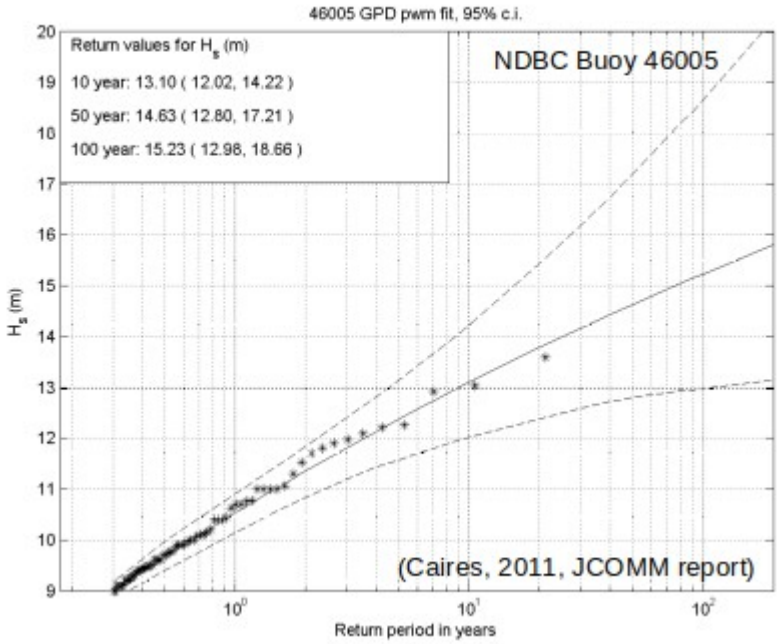


(Eckert-Gallup et al., 2016, *OE*)

Extreme wave design

1. Obtaining wave data at the desired location of sufficient duration

2. Applying extreme value theory and models to extrapolate to longer-term extremes



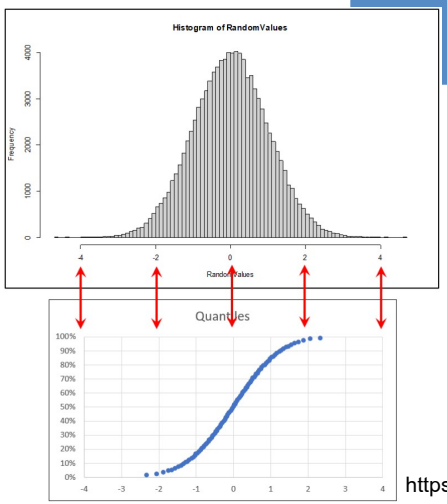
(Eckert-Gallup et al., 2016, *OE*)

Extreme wave design

1. Obtaining wave data at the desired location of sufficient duration

2. Applying extreme value theory and models to extrapolate to longer-term extremes

3. Estimating the Hs and Tp/Tm for a given return period



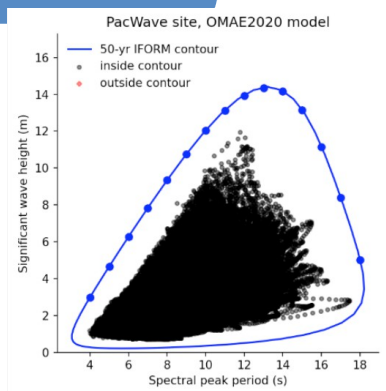
https://gsp.humboldt.edu/olm/R/02_03_qqplots.html

or

$$11,1\sqrt{\frac{H_{m0,OSS}}{g}} \leq T \leq 14,3\sqrt{\frac{H_{m0,OSS}}{g}}$$

Equation from: DTS-62600-2 (6.2.1.2)

or



(Eckert-Gallup et al., 2016, *OE*)

Extreme wave design

1. Obtaining wave data at the desired location of sufficient duration

2. Applying extreme value theory and models to extrapolate to longer-term extremes



3. Estimating the Hs and Tp/Tm for a given return period (independently or jointly)

4. Identifying one or more extreme states to use in a numerical or phyiscal model simulation

(Si et al., 2021, *OE*)

(Eckert-Gallup et al., 2016, *OE*)

Extreme value theory

Branch of statistics examining upper (lower) tails of probability distributions, assuming:

- Data independent and identically distributed
- Historical climate stationary, and future climate similar

Univariate analyses

- Block maxima approach (e.g. 1 year) + GEV (generalized extreme value) distribution
- Peaks-over-threshold (POT) + GPD (generalized Pareto distribution)
- R-largest values + r-largest distribution
- Initial-distribution method

(Caires, 2011, *JCOMM*)

Bivariate analyses

- Environmental contours approach:

(1) Joint probability density functions

$$f_{Hm0,Tp}(h, t) = f_{Hm0}(h)\, f_{Tp|Hm0}(t|h)$$

(2) Inverse First Order Reliability Method (IFORM)

(Hiles et al., 2019, *ECSS*)

Return period

- **Univariate: probability of exceedance**

$$\mathcal{T} = \frac{1}{\lambda \mathbb{P}[X > x]}$$

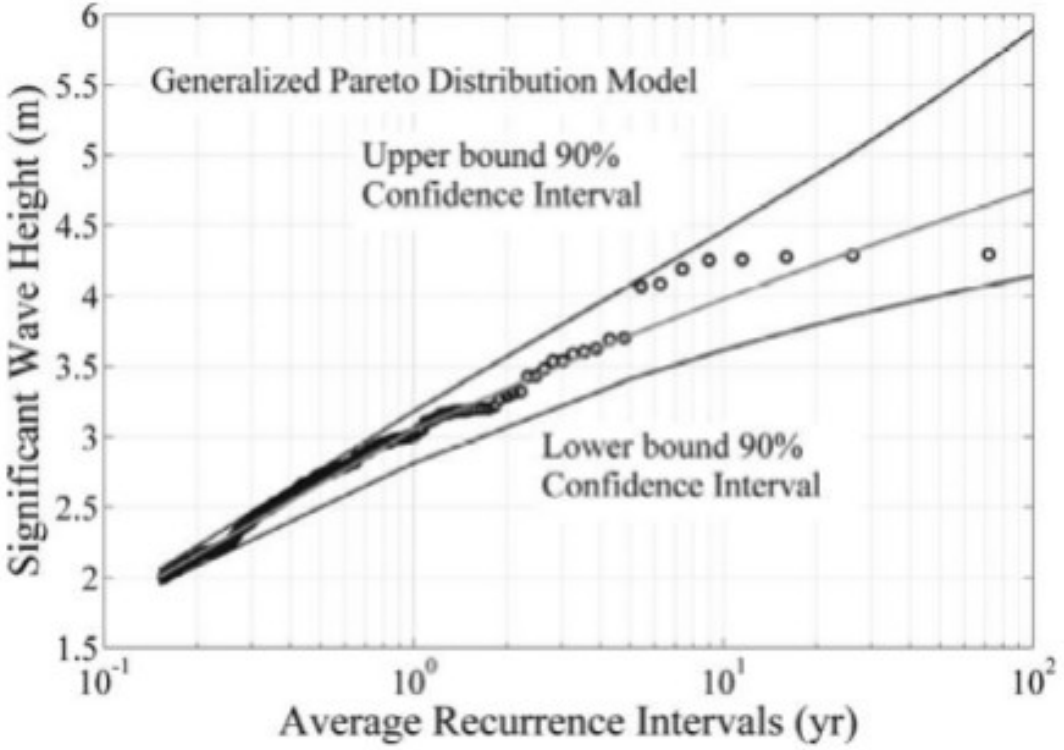
- **Bivariate:**

$$\mathcal{T}(x, y) = \frac{1}{\lambda_p \mathbb{P}[X > x, Y > y]}$$

Calculate return periods

Challenge: calculate *n*-year return period Hs (inverse of annual probability of occurrence)

- Often beyond the period of observations
- Extrapolating tails of extreme value distributions



Uncertainties?

(Far et al., 2018, *JCR*)

Characterization of extreme events: examples

- **Univariate:**

« GPD-Poisson model »

POT
« Peaks-Over-Threshold » declustering

+

GPD
« Generalized Pareto Distribution »

- **Bivariate:**

Copula analysis

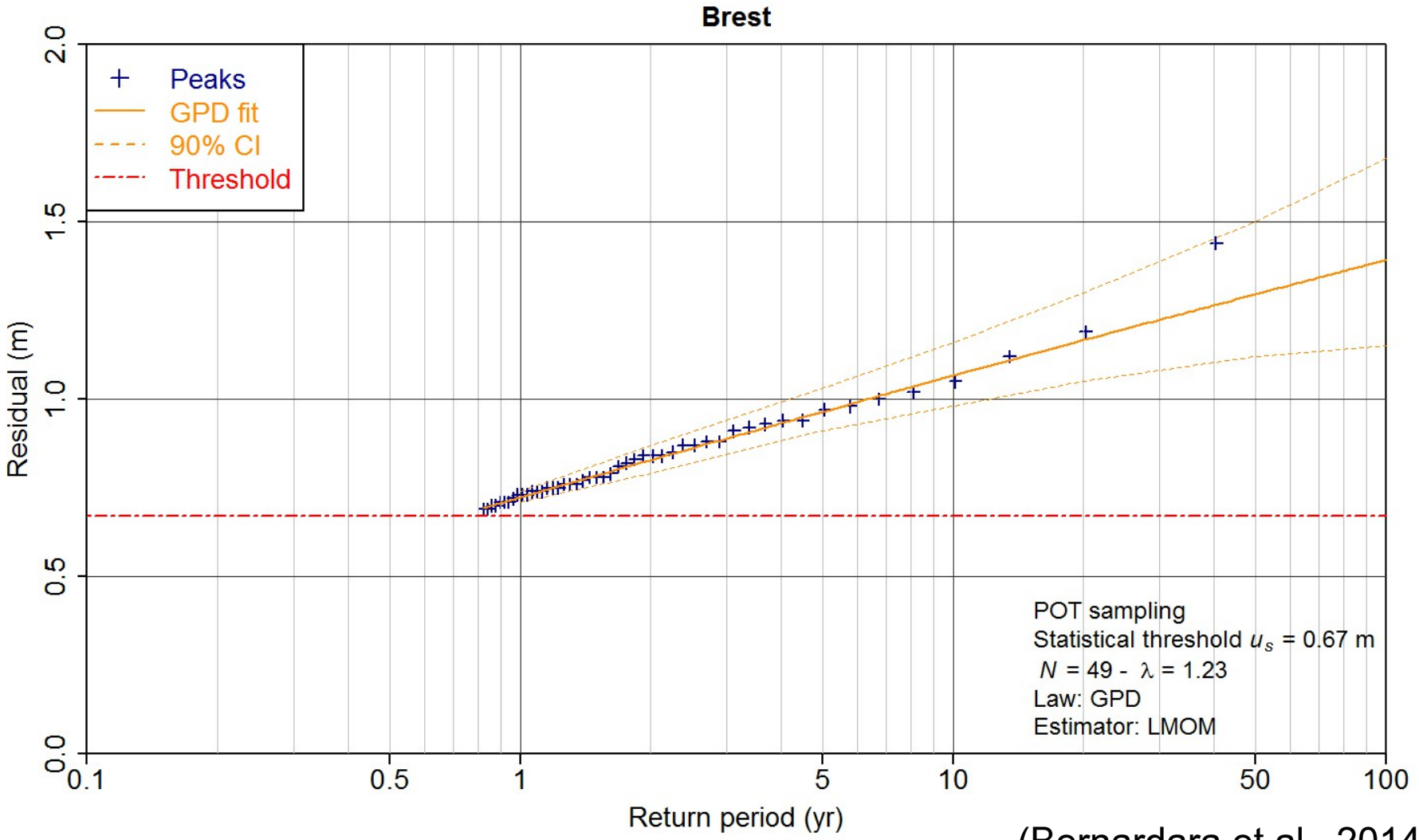
Data sampling
definition, identification, and description of event

+

Joint probability density function

Example: univariate analysis

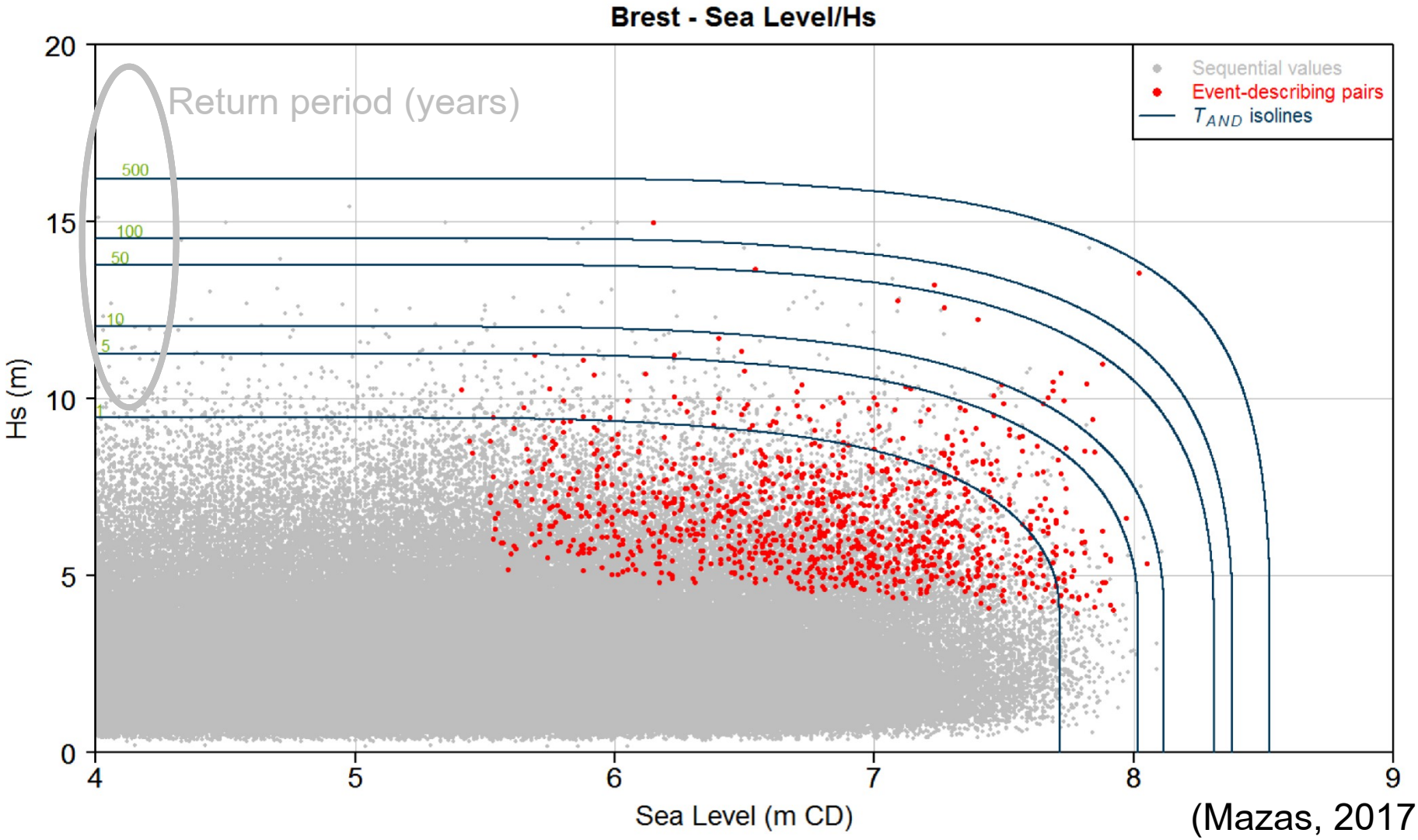
« GPD-Poisson model »



(Bernardara et al., 2014)

Example: bivariate analysis

Copula analysis



(Mazas, 2017)

Extreme value analysis (EVA): uncertainties

- **Choice of approach to use**

- **Free parameters in distribution fit**

- **Requires long-term data sets:**

  e.g. ISO standards for POT method recommends periods of record (POR) of a minimum of ¼ return period length (e.g. 12.5 years for 50 year return period) and 20 years for AM (annual maxima) methods

Univariate analyses

- Block maxima approach (e.g. 1 year) + GEV (generalized extreme value) distribution
- Peaks-over-threshold (POT) + GPD (generalized Pareto distribution)
- R-largest values + r-largest distribution
- Initial-distribution method

(Hiles et al., 2019, *ECSS*)

Extreme value analysis (EVA): uncertainties

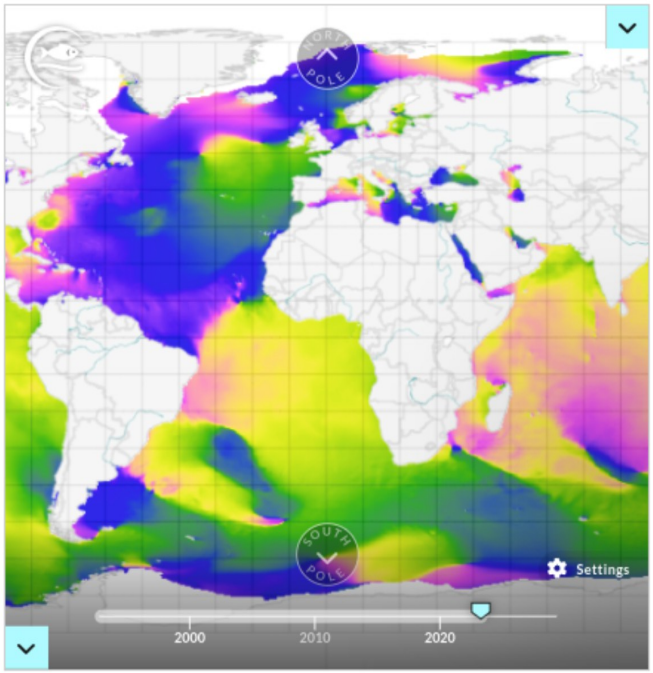
- **EVA using different data types:**
  - ***Observations (e.g. wave buoys):*** often underestimate extreme wave heights (limited spatial coverage and temporal duration)
  - ***Spectral wave model simulations:*** often underpredict large wave events and need observations for validation
- **Potentially large uncertainties:**
  - **Data limitations**
  - **Assuming stationarity**
  - **Nonstationarity uncertainties < EVA uncertainties** (in areas with slow changes, but extreme value stastitics change faster than mean value statistics)

(Neary et al., 2020, *JMSE*)

Wave analyses….

Availability of long-term wave simulations via Copernicus:

https://marine.copernicus.eu/

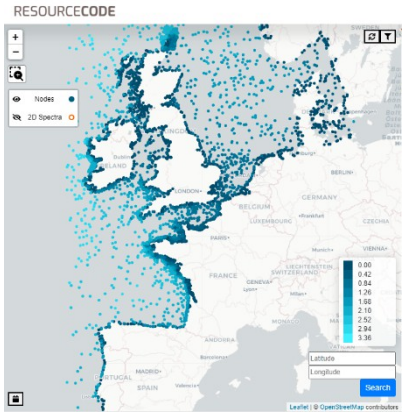


Today's class: extreme wave statistics

- How to estimate waves with a 1 and 50 year return period?
  - Adapt python code (project_wave_EVA.py) to estimate extreme Hs for your data set (must install pyextremes toolbox)
  - Then calculate 10, 50, and 90% quantile Tp

Wave analyses….

Availability of long-term wave simulations via RESOURCECODE (or another data base for your selected study site)



Today's class: extreme wave statistics

- How to estimate waves with a 1 and 50 year return period?
  - Adapt python code (project_wave_EVA.py) to estimate extreme Hs for your data set (must install pyextremes toolbox)
  - Then calculate 10, 50, and 90% quantile Tp

Wave analysis

Using the ResourceCode tools or the pyextremes toolbox…

- Calculate the extreme wave conditions:
  - 1 and 50 year return period Hs
  - 10, 50, and 90% quantile Tp associated with extreme Hs
  - Mean direction and directional spreading
- Estimate extreme wave height using the DNV-GL recommendations (approx.):
  - Estimate of 50-year max wave height:

    $H_{50yr} = 1.86\ H_{s,50y}$ (from mode of largest of 1000 Rayleigh-distributed wave heights)

    *factor may be up to 2.0 in deep water

Project schedule

Due dates in red: reports (R) and final presentation (FP)

I. Wave dynamics	Project	Lecturer	Date	
1. Introduction and characterizing ocean waves and sea states	**Description of class project and site selection**	M. Yates	25/09	
2. Linear wave theory	**Characterizing offshore waves**	M. Yates	09/10	
3. Nearshore wave propagation	**Calculating extreme wave conditions**	**M. Yates**	**16/10**	
4. Coastal hydrodyn			**06/11****	
II. Numerical wave				
5. Numerical wave r			**23/10****	**R1**
6. Numerical wave r			13/11	
III. Wave-structure				
7. Dynamics of a bo			20/11	
8. Modeling wave-b			**04/12**	**R2**
9. Offshore wind turbine foundations	**Final project work session**	C. Peyrard	11/12	
10. Final project presentations 14h - 18h15			***18/12***	**FP**

Last project work day for Part I : wave analyses

Individual write-ups due 23/10, max 10 pages
(due before midnight, or lose 1pt per day late)

To upload your completed report, please use the following link :
https://cloud.enpc.fr/s/B8MQJsigGxLM4ni